Madishetty Aditya Nath

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.32
Srichaitanya junior college	Class XII (Telangana Board of Intermediate Education)	2023	98.10%
Sree chaitanya high school	Class X (Board of Secondary Education, Telangana)	2021	10.00%

EXPERIENCE

Samsung R&D Institute India-Bangalore
Developer Intern

- **Automated Ticket Analysis:** Engineered an **AI-powered diagnosis system** for Samsung Ads DSP, reducing campaign troubleshooting from hours of manual investigation and Excel-based analysis to **23 seconds** with **96% ticket-match accuracy**, using hybrid semantic + keyword retrieval and **cross-encoder reranking** across 10,000+ support tickets.
- **Campaign Analysis Dashboard:** Developed a self-serve analytics platform for product managers, replacing hours of manual Excel analysis with **filter-based queries** and integrated **statistical and AI-driven analysis** for campaign performance insights across its lifecycle.

Research, Prof. Satya Peri
Research Student

- **Optimized** parallel Louvain community detection for large-scale graphs by reusing graph coloring across iterations and introducing **persistent thread-pool execution**, eliminating mutex contention and achieving up to **4× performance improvement** over the baseline implementation.

Teaching Assistant

Introduction to Software Programming (HT5023)

- **Guided** students through C and Python programming concepts, covering imperative, object-oriented, and scripting paradigms, and assisted with lab exercises on data types, loops, arrays, and functions.

OPEN SOURCE CONTRIBUTIONS

LLVM

Open Source Contributor (github)

- **Fortran PAUSE Support:** Collaborated with a co-author to extend Fortran **PAUSE** statement lowering in LLVM's Flang compiler, implementing **integer- and string-argument variants** alongside the existing no-argument case, with **control-flow-preserving codegen** in **C++** and new regression tests; **merged upstream**.

SELECTED PROJECTS

HISEN (Quantum Compiler)github

- Built a **C++** lexer and **Yacc/Bison** parser for a quantum programming language supporting functions, arrays, and **quantum primitives**.
- Designed a **AST hierarchy** with traversal utilities for structured representation of quantum programs.
- Implemented **symbol-table**-based **semantic analysis** for type, scope, and quantum-specific validation across test cases.

No-code-Form-Builder (github)

- Architected a versioned REST API with Node.js, Express, MongoDB, and Mongoose, using immutable form versions to enable safe iteration without affecting published forms.
- Built a recursive condition engine for nested AND/OR rules and 10+ operators, alongside AI-assisted form generation and RBAC-based workflow state transitions for submission approvals.

MCP Servers

- Implemented a **remote MCP** server exposing **4 YouTube tools** for subscription history, **personalized recommendations, subscription search**, and **library browsing** over **Streamable HTTP**, secured with **OAuth 2.1**. (github)
- Built a local MCP server with **9 Codeforces analysis tools** for activity analysis, problem recommendations, weak-topic detection, user comparison, contest tracking, and rating progress using the authenticated **Codeforces API**. (github)

Deep Learning Implementations (github)

- Implemented neural networks from scratch using **NumPy**, including **MLPs, CNNs, RNNs, LSTMs**, and **Transformer** encoders with manual forward/backward propagation.
- Built core deep-learning components including **BPTT, multi-head attention, optimizers, normalization, dropout, and gradient handling**, and evaluated models through architecture and optimization experiments.

Multiclient Quiz System (github)

- Built a multithreaded client-server quiz system in **C++** using raw **TCP** sockets, with cached and non-cached server and **LLM** generated questions .
- Designed a **custom HTTP-inspired application-layer protocol** for structured client-server communication over TCP.
- Analyzed network performance using **packet-capture tools**, measuring **throughput** and **latency** across multi-client scenarios.

SKILLS

- **Languages:** C++, Python, JavaScript, Golang*, RISC-V, HTML, CSS
- **Frameworks/Libraries:** React*, Node.js, Express, Tailwind CSS*, NumPy, Pandas, LangChain*
- **Tools and Platforms:** Git, GitHub, Render, Docker, LaTeX, Bash, Linux, MongoDB, MySQL
- **ML:** CNNs, PyTorch*, scikit-learn*, Deep Learning
- **CP Profiles :** Codeforces LeetCode

RELAVANT COURSES

- **Fundamentals:** Data Structures & Algorithms, Computer Networks, Operating Systems, Database Management Systems, Computer Architecture, Theory of Computation, Software Engineering,Theory of computation. compilers
- **Machine Learning:** Foundations of machine learning, Deep Learning, Linear algebra, convex optimization
- **Maths:** convex optimisations, linear algebra , probability and random variables
- **Electives:** Cryptology, Distributed computing, Quantum computing, Parallel and Concurrent Programming